
DAPHNE-SC Development

Moving board control from Linux-only service logic to Linux + RPU supervision

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The Sceptical Position

SHORT ANSWER

The RPU is a strong option only for local, measurable, fast protection. It is not a replacement for detector-wide knowledge.

- ▶ Good fit: bias current, rail current, board temperature, AFE power state.
- ▶ Bad fit: amount of light in the cryostat, who enabled a calibration system, global detector state.
- ▶ External permits can be inputs, but the RPU should not invent

Use the RPU for

Fast local shutdown when local electrical or thermal evidence says the board is unhealthy.

Do not claim

That the RPU understands optical exposure or global detector safety by itself.



SECTION

Main Question

Which Linux failure modes can the RPU catch locally, and which ones remain outside the board?



Main Slide: Failure Modes Covered By The RPU

Linux hangs or restarts

RPU sees stale Linux heartbeat but still samples bias current, rail current, and temperature. It can trip both bias enables or AFE power on a local fault.

Network / OPC-UA is down

Operators lose visibility, but the RPU keeps the local protection loop alive and refuses new remote authority.

Linux polling is stale

RPU uses fresh local samples, sample-age checks, debounce, and hysteresis. It latches a fault if samples are stale or unhealthy.

Bad enable after fault

RPU keeps the fault latched, rejects re-enable, and requires explicit clear plus controlled recovery.

Sceptical boundary

This only works if the RPU has an independent monitor path and a real actuator path to remove bias or AFE power. Otherwise it is just another alarm.

Concrete Local Trip Cases

Bias current

Switch off both SiPM biases when current leaves the healthy window and persists through debounce.

Temperature

Remove bias and/or AFE power when board, regulator, or AFE temperature crosses the trip threshold.

Rail current

Disable AFE or bias rail when current/power indicates an electrical fault.

Why RPU instead of Linux

The RPU can keep running a small finite-state machine while Linux restarts, the network is down, the OPC-UA bridge is stale, or DAQ is not responding.

Sceptical limit

This is not a physics trigger or a light monitor. It reacts to local electrical and thermal consequences, not to the original global cause.

What The RPU Cannot Know Locally

Not local information

- ▶ light level inside the cryostat;
- ▶ laser, LCM, or flasher state unless wired in;
- ▶ cryogenics, rack, or detector-wide permit state unless wired in;
- ▶ whether high SiPM current is physics light, noise, damage, or a control mistake.

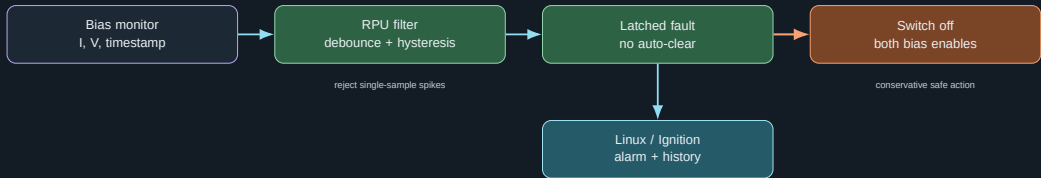
Local information

- ▶ bias voltage and bias current;
- ▶ local rail current and voltage;
- ▶ local temperature;
- ▶ watchdogs, stale samples, and fault latches;
- ▶ external permit bits if they are provided as inputs.

Therefore

The honest claim is narrower: the RPU protects against locally observable electrical and thermal unhealthy states.

Example: Bias Current Guard



Trip condition

Current outside the healthy SiPM range for long enough to survive debounce and threshold hysteresis.

Hard question

Do we trust the local current monitor enough to remove bias without Linux confirmation?

Should It Switch Off Both Biases?

Argument for both

- ▶ common-cause fault is possible;
- ▶ mapping may be ambiguous during commissioning;
- ▶ detector protection beats availability;
- ▶ a single bad bias path may indicate a shared supply or board problem.

Argument against both

- ▶ unnecessary detector dead region if the fault is isolated;
- ▶ false trip rate matters during operations;
- ▶ one threshold may not fit all temperature and SiPM conditions;
- ▶ recovery becomes slower and more disruptive.

Conservative proposal

Start with both-bias shutdown for high-confidence, persistent, class-A faults. Keep per-bias shutdown as a later refinement only if monitors and mapping are proven.

AFE Shutdown Is A Different Decision

Good RPU case

AFE rail current, AFE power, or local temperature is high enough that continued operation could damage hardware.

Local action

Disable AFE power or place the frontend in a defined safe state, then latch the reason.

Do not overreach

Do not turn off AFEs because the RPU suspects too much light. It cannot know that locally unless the light system permit is wired in.

Escalation

If the required action must survive board firmware or RPU failure, it belongs in DPS or a supply-side inhibit.

The Algorithm Must Be Boring

For protection, boring is a feature: thresholds, timers, latches, explicit reset, and testable failure behavior.

What The Local Algorithm Should Be

Allowed

- ▶ calibrated thresholds;
- ▶ hysteresis between trip and release;
- ▶ debounce time for noisy ADC samples;
- ▶ stale-sample detection;
- ▶ latched trip with explicit reset;
- ▶ no automatic re-enable of bias.

Not allowed

- ▶ complex diagnosis of light source;
- ▶ ML or adaptive policy in the protection loop;
- ▶ hidden threshold changes;
- ▶ recovery without operator or DAQ revalidation;
- ▶ silent failure to a permissive state.

What Must Be Proven Before We Trust It

Question	Required evidence
Can the RPU really switch off bias?	Hardware path from RPU to bias enable or supply inhibit, tested under load.
Are current and temperature measurements trustworthy?	Calibration, sample rate, stale-sample detection, and fault-injection tests.
Are thresholds physically meaningful?	Healthy SiPM current ranges versus temperature, bias setting, noise, and expected operating state.
What is the false-trip cost?	Operations impact if bias is removed during a good run.
	Damage or detector-risk analysis if the RPU waits too long or does nothing.

Recommended Decision Language

SAY THIS

We propose the RPU as a local protection layer for DAPHNE electrical and thermal faults, especially SiPM bias-current trips and AFE/rail thermal or over-current faults.

- ▶ It reacts locally when Linux or the network is unavailable.
- ▶ It latches faults and requires controlled recovery.
- ▶ It reports to Linux, Ignition, and DAQ after the fact.

Do not say

The RPU protects against unknown light in the cryostat unless a light-system permit is wired to it.

Do say

Detector-wide or hard protection still belongs to DPS or supply-side inhibit when independence is required.

Decision Gate

Proceed if

Local bias current, rail current, and temperature monitors are reliable enough to justify automatic local shutdown.

Proceed if

The RPU output path can remove both bias enables and, for thermal/electrical faults, place AFEs in a safe state.

Do not proceed if

Thresholds are guesses, monitor freshness is not guaranteed, or recovery behavior is not defined.

Do not proceed if

The action must be independent of board software but is not wired through DPS or a supply-side inhibit.

Takeaway

BOTTOM LINE

Yes, it makes sense, but only with a narrow claim: the RPU is a critical local guard for electrical and thermal health, not a detector-wide protection oracle.

Bias current trip, temperature trip, rail-current trip, latch, report, controlled reset.

Best RPU case

Switch off both SiPM biases when local current is persistently outside the healthy range.

Still needed

DPS or supply inhibit for protection that must survive board-software failure.